SG No. 214 O^8 $I4_132$ [cubic]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$

Table 1: Wyckoff site: `8a`, site symmetry: `.32`

No.	position	mapping
1	$\left[\frac{1}{8},\ \frac{1}{8},\ \frac{1}{8}\right]$	`[1,5,9,14,19,24]`
2	$\left[\frac{3}{8},\ \frac{7}{8},\ \frac{5}{8}\right]$	`[2,7,12,13,17,21]`
3	$\left[\frac{7}{8},\ \frac{5}{8},\ \frac{3}{8}\right]$	`[3,8,10,15,20,22]`
4	$\left[\frac{5}{8},\ \frac{3}{8},\ \frac{7}{8}\right]$	`[4,6,11,16,18,23]`

Table 2: Wyckoff site: `8b`, site symmetry: `.32`

No.	position	mapping
1	$\left[\frac{7}{8},\ \frac{7}{8},\ \frac{7}{8}\right]$	`[1,5,9,14,19,24]`
2	$\left[\frac{5}{8},\ \frac{1}{8},\ \frac{3}{8}\right]$	`[2,7,12,13,17,21]`
3	$\left[\frac{1}{8},\ \frac{3}{8},\ \frac{5}{8}\right]$	`[3,8,10,15,20,22]`
4	$\left[\frac{3}{8},\ \frac{5}{8},\ \frac{1}{8}\right]$	`[4,6,11,16,18,23]`

Table 3: Wyckoff site: `12c`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{1}{8},\ 0,\ \frac{1}{4}\right]$	`[1,4,18,19]`
2	$\left[\frac{3}{8},\ 0,\ \frac{3}{4}\right]$	`[2,3,17,20]`
3	$\left[\frac{1}{4},\ \frac{1}{8},\ 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4},\ \frac{3}{8},\ 0\right]$	`[6,7,13,16]`
5	$\left[0,\ \frac{1}{4},\ \frac{1}{8}\right]$	`[9,12,21,24]`
6	$\left[0,\ \frac{3}{4},\ \frac{3}{8}\right]$	`[10,11,22,23]`

Table 4: Wyckoff site: `12d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[\frac{5}{8},\ 0,\ \frac{1}{4}\right]$	`[1,4,18,19]`
2	$\left[\frac{7}{8},\ 0,\ \frac{3}{4}\right]$	`[2,3,17,20]`
3	$\left[\frac{1}{4},\ \frac{5}{8},\ 0\right]$	`[5,8,14,15]`
4	$\left[\frac{3}{4},\ \frac{7}{8},\ 0\right]$	`[6,7,13,16]`
5	$\left[0,\ \frac{1}{4},\ \frac{5}{8}\right]$	`[9,12,21,24]`
6	$\left[0,\ \frac{3}{4},\ \frac{7}{8}\right]$	`[10,11,22,23]`

Table 5: Wyckoff site: `16e`, site symmetry: `.3.`

No.	position	mapping
1	$[x,\ x,\ x]$	`[1,5,9]`
2	$\left[\frac{1}{2}-x,\ -x,\ x+\frac{1}{2}\right]$	`[2,7,12]`
3	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[3,8,10]`

continued ...

Table 5

No.	position	mapping
4	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -x\right]$	[4,6,11]
5	$\left[x+\frac{3}{4},\ x+\frac{1}{4},\ \frac{1}{4}-x\right]$	[13,17,21]
6	$\left[\frac{3}{4}-x,\ \frac{3}{4}-x,\ \frac{3}{4}-x\right]$	[14,19,24]
7	$\left[x+\frac{1}{4},\ \frac{1}{4}-x,\ x+\frac{3}{4}\right]$	[15,20,22]
8	$\left[\frac{1}{4}-x,\ x+\frac{3}{4},\ x+\frac{1}{4}\right]$	[16,18,23]

Table 6: Wyckoff site: 24f, site symmetry: 2..

No.	position	mapping
1	$\left[x,\ 0,\ \frac{1}{4}\right]$	[1,4]
2	$\left[\frac{1}{2}-x,\ 0,\ \frac{3}{4}\right]$	[2,3]
3	$\left[\frac{1}{4},\ x,\ 0\right]$	[5,8]
4	$\left[\frac{3}{4},\ \frac{1}{2}-x,\ 0\right]$	[6,7]
5	$\left[0,\ \frac{1}{4},\ x\right]$	[9,12]
6	$\left[0,\ \frac{3}{4},\ \frac{1}{2}-x\right]$	[10,11]
7	$\left[\frac{3}{4},\ x+\frac{1}{4},\ 0\right]$	[13,16]
8	$\left[\frac{3}{4},\ \frac{3}{4}-x,\ \frac{1}{2}\right]$	[14,15]
9	$\left[x+\frac{3}{4},\ \frac{1}{2},\ \frac{1}{4}\right]$	[17,20]
10	$\left[\frac{1}{4}-x,\ 0,\ \frac{1}{4}\right]$	[18,19]
11	$\left[0,\ \frac{1}{4},\ \frac{1}{4}-x\right]$	[21,24]
12	$\left[\frac{1}{2},\ \frac{1}{4},\ x+\frac{3}{4}\right]$	[22,23]

Table 7: Wyckoff site: 24g, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{8},\ y,\ y+\frac{1}{4}\right]$	[1,18]
2	$\left[\frac{3}{8},\ -y,\ y+\frac{3}{4}\right]$	[2,20]
3	$\left[\frac{7}{8},\ y+\frac{1}{2},\ \frac{1}{4}-y\right]$	[3,17]
4	$\left[\frac{5}{8},\ \frac{1}{2}-y,\ \frac{3}{4}-y\right]$	[4,19]
5	$\left[y+\frac{1}{4},\ \frac{1}{8},\ y\right]$	[5,15]
6	$\left[y+\frac{3}{4},\ \frac{3}{8},\ -y\right]$	[6,13]
7	$\left[\frac{1}{4}-y,\ \frac{7}{8},\ y+\frac{1}{2}\right]$	[7,16]
8	$\left[\frac{3}{4}-y,\ \frac{5}{8},\ \frac{1}{2}-y\right]$	[8,14]
9	$\left[y,\ y+\frac{1}{4},\ \frac{1}{8}\right]$	[9,21]
10	$\left[-y,\ y+\frac{3}{4},\ \frac{3}{8}\right]$	[10,23]
11	$\left[y+\frac{1}{2},\ \frac{1}{4}-y,\ \frac{7}{8}\right]$	[11,22]
12	$\left[\frac{1}{2}-y,\ \frac{3}{4}-y,\ \frac{5}{8}\right]$	[12,24]

Table 8: Wyckoff site: 24h, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{8},\ y,\ \frac{1}{4}-y\right]$	[1,19]
2	$\left[\frac{3}{8},\ -y,\ \frac{3}{4}-y\right]$	[2,17]

continued ...

Table 8

No.	position	mapping
3	$\left[\frac{7}{8}, y+\frac{1}{2}, y+\frac{1}{4}\right]$	[3,20]
4	$\left[\frac{5}{8}, \frac{1}{2}-y, y+\frac{3}{4}\right]$	[4,18]
5	$\left[\frac{1}{4}-y, \frac{1}{8}, y\right]$	[5,14]
6	$\left[\frac{3}{4}-y, \frac{3}{8}, -y\right]$	[6,16]
7	$\left[y+\frac{1}{4}, \frac{7}{8}, y+\frac{1}{2}\right]$	[7,13]
8	$\left[y+\frac{3}{4}, \frac{5}{8}, \frac{1}{2}-y\right]$	[8,15]
9	$\left[y, \frac{1}{4}-y, \frac{1}{8}\right]$	[9,24]
10	$\left[-y, \frac{3}{4}-y, \frac{3}{8}\right]$	[10,22]
11	$\left[y+\frac{1}{2}, y+\frac{1}{4}, \frac{7}{8}\right]$	[11,23]
12	$\left[\frac{1}{2}-y, y+\frac{3}{4}, \frac{5}{8}\right]$	[12,21]

Table 9: Wyckoff site: 48i, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$	[2]
3	$\left[-x, y+\frac{1}{2}, \frac{1}{2}-z\right]$	[3]
4	$\left[x+\frac{1}{2}, \frac{1}{2}-y, -z\right]$	[4]
5	$[z, x, y]$	[5]
6	$\left[z+\frac{1}{2}, \frac{1}{2}-x, -y\right]$	[6]
7	$\left[\frac{1}{2}-z, -x, y+\frac{1}{2}\right]$	[7]
8	$\left[-z, x+\frac{1}{2}, \frac{1}{2}-y\right]$	[8]
9	$[y, z, x]$	[9]
10	$\left[-y, z+\frac{1}{2}, \frac{1}{2}-x\right]$	[10]
11	$\left[y+\frac{1}{2}, \frac{1}{2}-z, -x\right]$	[11]
12	$\left[\frac{1}{2}-y, -z, x+\frac{1}{2}\right]$	[12]
13	$\left[y+\frac{3}{4}, x+\frac{1}{4}, \frac{1}{4}-z\right]$	[13]
14	$\left[\frac{3}{4}-y, \frac{3}{4}-x, \frac{3}{4}-z\right]$	[14]
15	$\left[y+\frac{1}{4}, \frac{1}{4}-x, z+\frac{3}{4}\right]$	[15]
16	$\left[\frac{1}{4}-y, x+\frac{3}{4}, z+\frac{1}{4}\right]$	[16]
17	$\left[x+\frac{3}{4}, z+\frac{1}{4}, \frac{1}{4}-y\right]$	[17]
18	$\left[\frac{1}{4}-x, z+\frac{3}{4}, y+\frac{1}{4}\right]$	[18]
19	$\left[\frac{3}{4}-x, \frac{3}{4}-z, \frac{3}{4}-y\right]$	[19]
20	$\left[x+\frac{1}{4}, \frac{1}{4}-z, y+\frac{3}{4}\right]$	[20]
21	$\left[z+\frac{3}{4}, y+\frac{1}{4}, \frac{1}{4}-x\right]$	[21]
22	$\left[z+\frac{1}{4}, \frac{1}{4}-y, x+\frac{3}{4}\right]$	[22]
23	$\left[\frac{1}{4}-z, y+\frac{3}{4}, x+\frac{1}{4}\right]$	[23]
24	$\left[\frac{3}{4}-z, \frac{3}{4}-y, \frac{3}{4}-x\right]$	[24]